DC GENERATORS / DC MOTORS

DC Machines

- DC Generator: Mechanical energy → Electrical energy
- DC Motor: Electrical energy → Mechanical energy

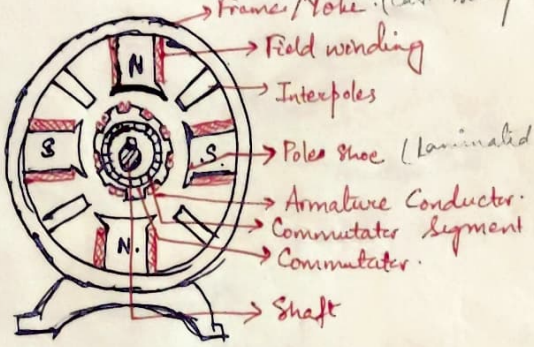
The DC Generator is based on the principle that when a conductor is rotated in a dc magnetic field, a voltage will be generated in the conductor.

DC Generator Construction:

A dc generator has three main parts —

1. Magnetic field system
2. Armature
3. Commutator & brush gear

Stator { Yoke, Poles, field coil
Rotor → Armature, Commutator



1. Magnetic Field System

→ It is stationary or fixed part of the machine.
→ It produces the main magnetic flux.

- Yoke (outer frame)
  1. Supports the pole cores
  2. Protecting cover to the m/c
  3. forms a part of the magnetic circuit.
- Pole core (in even nos).
- Pole Shoe
  - ↳ curved shoe surface
    1. supports the field coil
    2. C/s area ↑ ⇒ Reluctance ↓

→ Yoke → cylinder of cast steel or rolled steel.
→ Pole Core → laminated steel sheets. (cast steel)
→ Since poles project inwards, they are called salient poles.
→ Field coils connected in series such that when the current flows through the coil, alternate N and S poles are produced.

Armature

→ The rotating part of the dc machine is called armature.

→ Shaft → Armature Core - (Slots) - Armature wdgs.

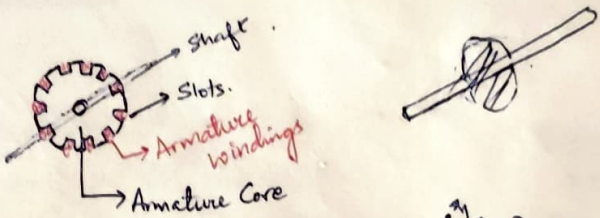
→ The armature consists of shaft upon which laminated cylinder called armature core is mounted.

→ The armature core has slots / grovers on its outer surface in which insulated conductors (armature wdg) are present.

→ The conductors are wedged and band of steel wire are fastened round the core to prevent them flying under the centrifugal forces.

→ The conductors are suitably connected. This connected arrangement of conductors is called armature winding.

→ Two types of windings are used - lap and wave.



Waveform of emf generated is

Commutator and Brushgear.

→ Why commutator is required?
Alternating voltage is produced in a coil rotating in a magnetic field. To obtain direct current in the external circuit, a commutator is needed.

→ Commutator rotates with the armature and is made up of Cu bars.

→ Dc Current is collected from the armature winding by means of two or more carbon brushes mounted on the commutator.

Functions of commutator:
* Commutator facilitates the collection of current from the armature conductors.
* It rectifies ie converts the ac induced current in the armature into unidirectional current.
* Brushes collect current from commutator.

Magnetic Circuit of a DC Generator:

0.5 mm to 1.5 mm of air gap.

→ The air gap is the space between the armature surface & the pole face. This space is kept as small as possible.